Comparison of Fine-Tuning Paradigms for Diagnostic SLMs

SLM for Disease Diagnosis Project

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Table 1: **Fine-Tuning WITHOUT RAG vs. Fine-Tuning WITH RAG (RAFT)**

Dimension	**Fine-Tuning WITHOUT RAG (Closed-Book)**	**Fine-Tuning WITH RAG (RAFT / Open-Book)**
Knowledge Source	Internal model weights (Parametric)	External Database / Knowledge Graph (Non-parametric)
3B Model Capacity	Bottlenecked; parameter interference	Unconstrained; weights focus purely on reasoning
Hallucination Risk	High (unverifiable memory recall)	Low (strictly grounded in retrieved evidence)
Noise / Distractors	None (no external context)	High resistance (trained on hard negative distractors)
Clinical Audit	Black-box justification	Verifiable citations to guideline rules & patient twins
Guideline Updates	Requires full model retraining	Instant (update external database index only)
Input Token Length	Short ($\sim$200 tokens)	Long ($\sim$2,000–3,000 tokens)
Inference Latency	Ultra-fast ($\sim$15–30 ms)	Moderate ($\sim$150–250 ms)
System Footprint	Standalone SLM binary (Edge-friendly)	SLM + Retriever + Vector DB / Graph